

Midterm Assessment 1

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DATA-581

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1 1. Executive Summary (20 points)

This was primarily a learning and exploratory investigation. The data was probed for general correlations that would be interesting to further investigate. There are some simple numerical representations and some useful graphical representations after investigations progressed. The investigation revealed various combinations of species across the three islands with Adell across all three islands but Chinstrap and Gentoo penguins were present on only one island. The investigation also revealed that Gentoo penguins are statistically larger than Chinstrap and Adell which were of comparable mass on average.

2 2. Exploratory Data Analysis

Below are the exploratory techniques used to learn about this dataset

2.1 Load and Clean Data

Load penguins dataset. Drop all rows that have missing values. 344 total rows in the raw data but 333 rows in the cleaned data dropping rows with null values in species, island, and sex.

```
[174]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[176]: # Load Data
penguins_raw = sns.load_dataset("penguins")
#raw_count = penguins_raw.count()
raw_count = len(penguins_raw)
```

```
# Clean data
penguins = penguins_raw.dropna(subset=['species', 'island', 'sex'])
clean_count = len(penguins)

print("Raw Dataframe Count: {}".format(raw_count))
print("Clean Dataframe Count: {}".format(clean_count))
```

Raw Dataframe Count: 344
Clean Dataframe Count: 333

2.2 Summary Statistics

Below are the numerical explorations done to learn about the data

```
[177]: def line_print():
        print("="*40)

float_field_keys = [
    ("Bill Length(mm)", "bill_length_mm",),
    ("Bill Depth(mm)", "bill_depth_mm",),
    ("Flipper Length(mm)", "flipper_length_mm",),
    ("Body Mass(grams)", "body_mass_g",),
]

species_group = penguins.groupby("species")

math_ops = [
    "mean",
    "median",
    "std",
    "var",
    "min",
    "max",
]

for pretty_name, key in float_field_keys:
    line_print()
    line_print()
    print("Field: {}".format(pretty_name))
    line_print()
    print(species_group[key].agg(math_ops))
    line_print()

print()
for i in range(0,4):
    line_print()
print()
```

```

other_keys = [
    ("Island", "island",),
    ("Species", "species",),
    #("", "",),
    #("", "",),
]

for name, key in other_keys:
    group_by_columns = ["sex"]
    group_by_columns.append(key)
    grouped_penguins = penguins.groupby(group_by_columns)
    line_print()
    print("Sex grouped by {}".format(pretty_name))
    line_print()
    print(grouped_penguins.count())
    line_print()

```

```

=====
=====
Field: Bill Length(mm)
=====

```

	mean	median	std	var	min	max
species						
Adelie	38.823973	38.85	2.662597	7.089421	32.1	46.0
Chinstrap	48.833824	49.55	3.339256	11.150630	40.9	58.0
Gentoo	47.568067	47.40	3.106116	9.647955	40.9	59.6

```

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```

```

Field: Bill Depth(mm)
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```

	mean	median	std	var	min	max
species						
Adelie	18.347260	18.40	1.219338	1.486786	15.5	21.5
Chinstrap	18.420588	18.45	1.135395	1.289122	16.4	20.8
Gentoo	14.996639	15.00	0.985998	0.972192	13.1	17.3

```

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```

```

Field: Flipper Length(mm)
=====

```

	mean	median	std	var	min	max
species						
Adelie	190.102740	190.0	6.521825	42.534199	172.0	210.0
Chinstrap	195.823529	196.0	7.131894	50.863916	178.0	212.0
Gentoo	217.235294	216.0	6.585431	43.367896	203.0	231.0

```
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```

Field: Body Mass(grams)

```
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```

	mean	median	std	var	min	max
species						
Adelie	3706.164384	3700.0	458.620135	210332.427964	2850.0	4775.0
Chinstrap	3733.088235	3700.0	384.335081	147713.454785	2700.0	4800.0
Gentoo	5092.436975	5050.0	501.476154	251478.332859	3950.0	6300.0

```
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Sex grouped by Body Mass(grams)

```
=====
```

	species	bill_length_mm	bill_depth_mm	flipper_length_mm	\
sex	island				
Female	Biscoe	80	80	80	80
	Dream	61	61	61	61
	Torgersen	24	24	24	24
Male	Biscoe	83	83	83	83
	Dream	62	62	62	62
	Torgersen	23	23	23	23

	body_mass_g
sex	island
Female	Biscoe
	Dream
	Torgersen
Male	Biscoe
	Dream
	Torgersen

```
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```

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```

Sex grouped by Body Mass(grams)

```
=====
```

	island	bill_length_mm	bill_depth_mm	flipper_length_mm	\
sex	species				
Female	Adelie	73	73	73	73
	Chinstrap	34	34	34	34
	Gentoo	58	58	58	58
Male	Adelie	73	73	73	73
	Chinstrap	34	34	34	34

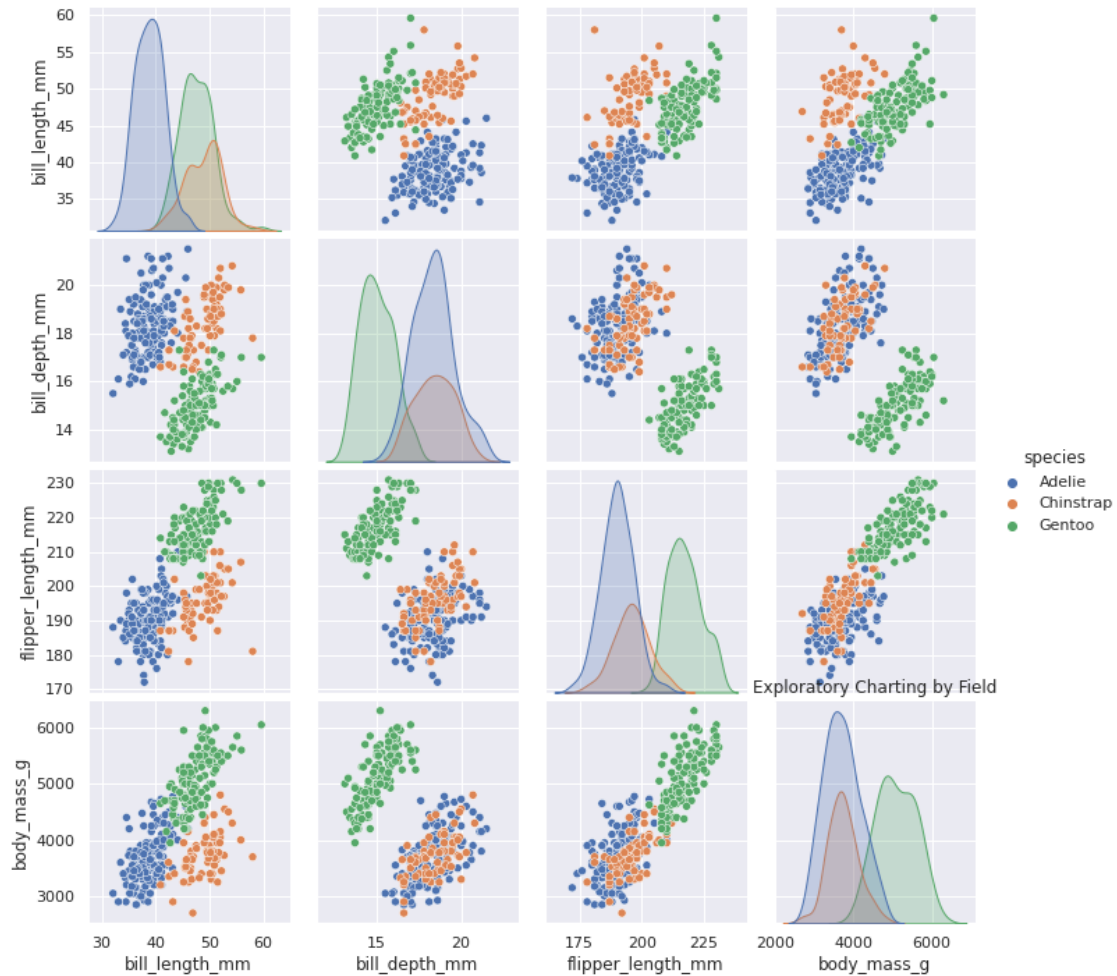
	sex	species	body_mass_g
	Female	Adelie	73
		Chinstrap	34
		Gentoo	58
	Male	Adelie	73
		Chinstrap	34
		Gentoo	61

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2.3 Categorical Comparison

Here is a large categorical comparison between all the different fields in the penguins dataset. I'm trying to visualize different relationships between the various fields based on species types as well which is the hue.

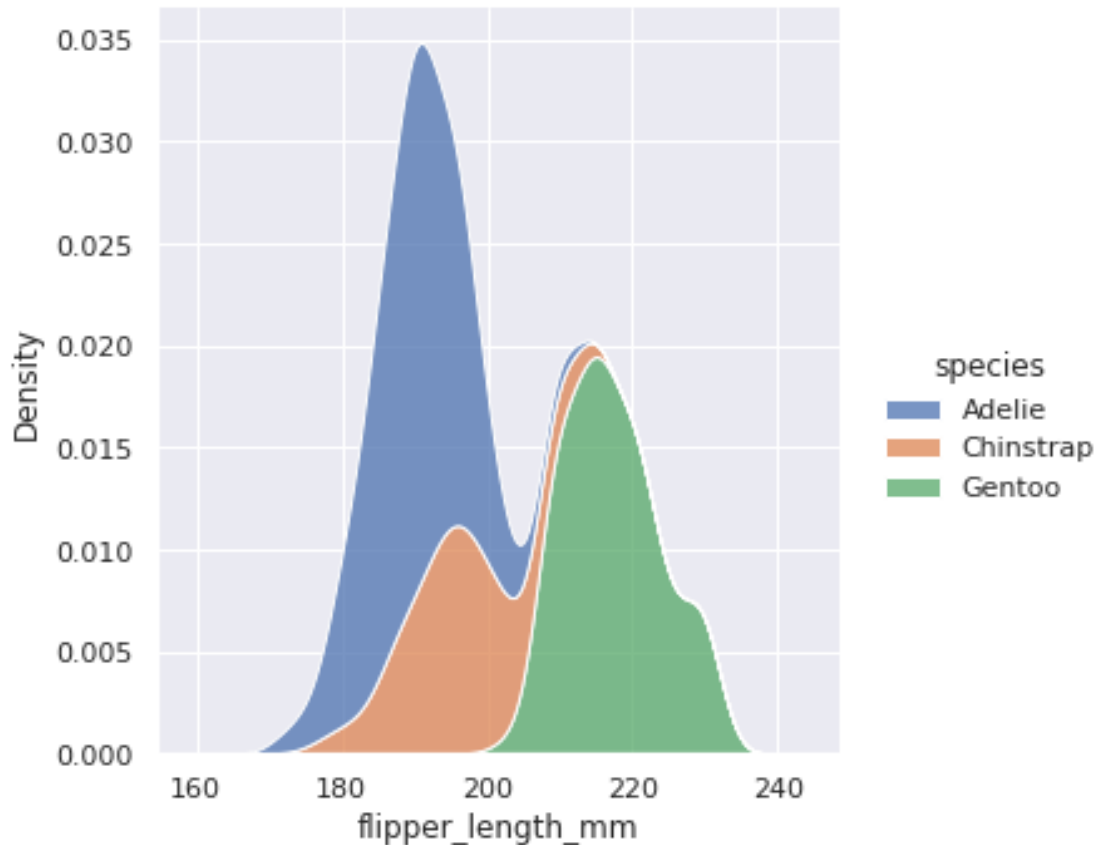
```
[178]: sns.pairplot(penguins, hue="species")
plt.title('Exploratory Charting by Field')
plt.show()
```



2.4 Distribution chart

Here is a distribution chart showing the distribution of Flipper Length across each species. Adelle have the widest distribution with the largest total population while Gentoo have the smallest distribution but have the largest flippers on average.

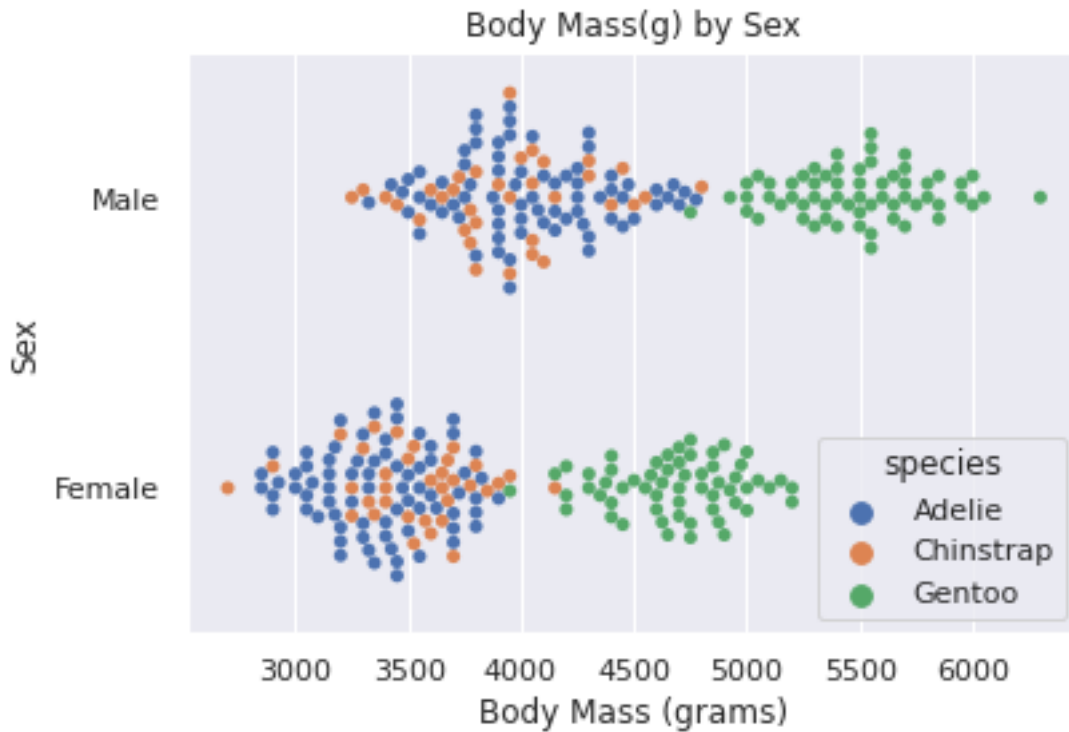
```
[179]: sns.displot(data=penguins, x="flipper_length_mm", hue="species",
    ↪multiple="stack", kind="kde")
plt.show()
```



2.5 Relationship Chart

Here is a chart showing the relations between penguin mass by sex across each population. In it you can see that Gentoo have the larger penguins in each sex while the Adell and Chinstrap are approximately the same size.

```
[180]: ax = sns.swarmplot(data=penguins, x="body_mass_g", y="sex", hue="species")
ax.set(ylabel="Sex", xlabel="Body Mass (grams)")
plt.title('Body Mass(g) by Sex')
plt.show()
```

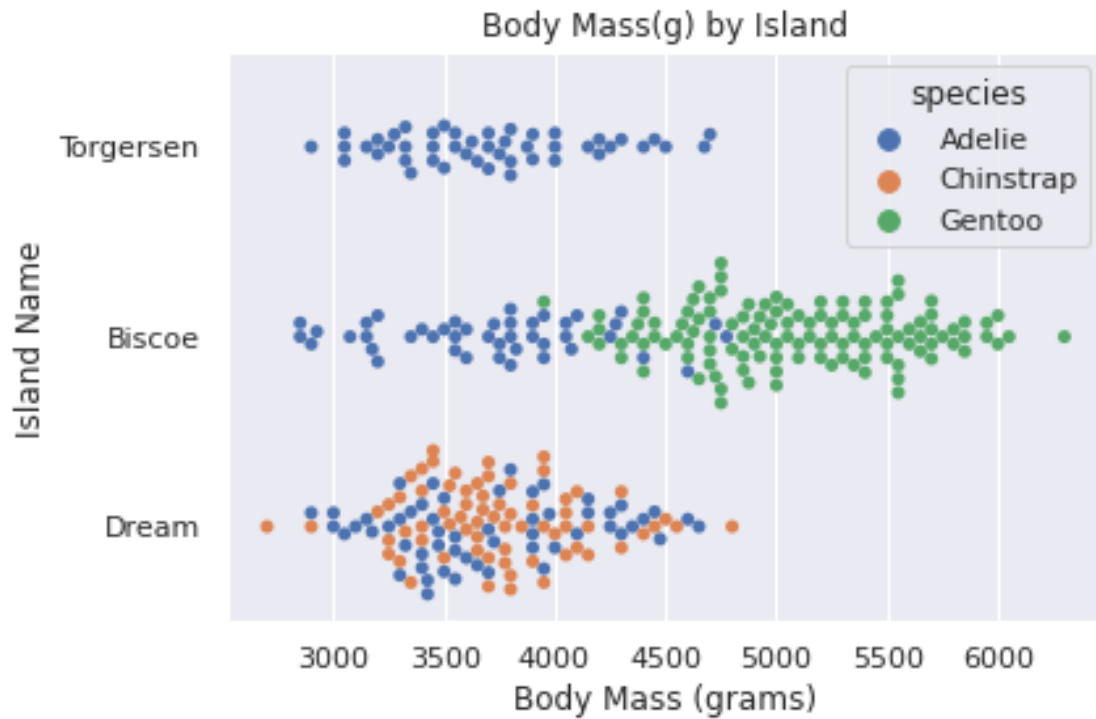


2.6 Other Interesting Graphical Representations

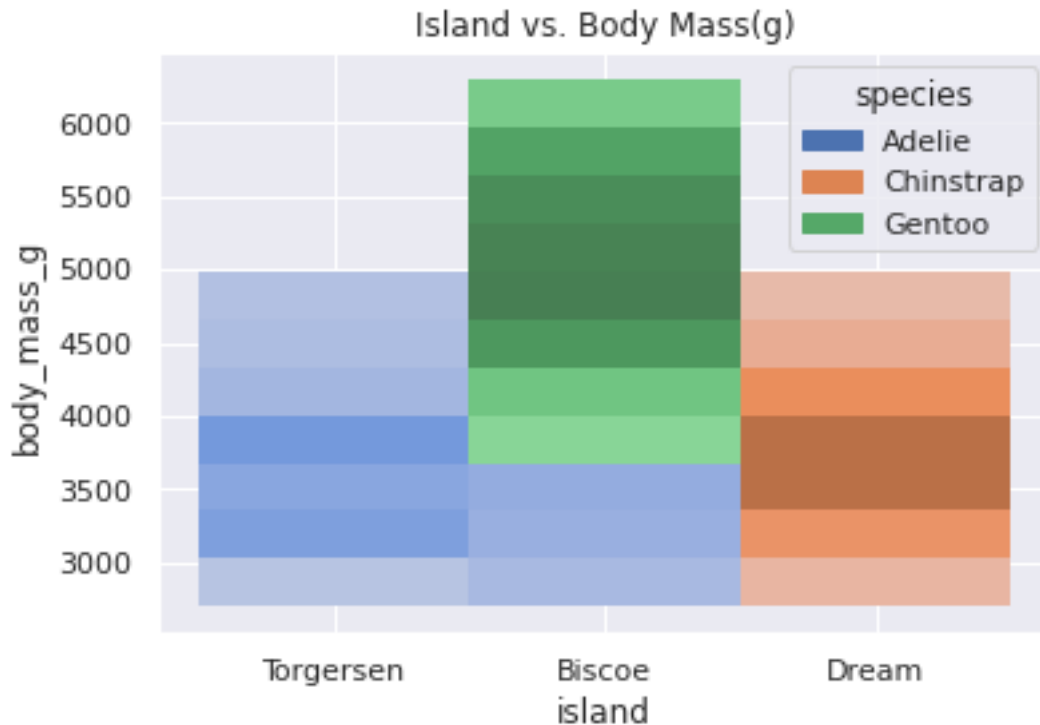
Below is swarm plot and a histogram plot with penguin mass grouped by sex and colored based on species.

In this data, it shows that Adelle penguins are on all three islands while Gentoo is only on Biscoe and Chinstrap are only on Dream.

```
[181]: # Draw a categorical scatterplot to show each observation
ax = sns.swarmplot(data=penguins, x="body_mass_g", y="island", hue="species")
ax.set(ylabel="Island Name", xlabel="Body Mass (grams)")
plt.title('Body Mass(g) by Island')
plt.show()
```

```
[182]: sns.histplot(x='island', y='body_mass_g', hue='species', data=penguins,
↳ multiple="stack")
plt.title('Island vs. Body Mass(g)')
ax.set(ylabel="Island Name", xlabel="Body Mass (grams)")
plt.show()
```



3 3. Ethics Concerns

3.0.1 How can data visualizations unintentionally mislead audiences?

It can lead them to conclusions that aren't true. As saying goes, there are lies, damn lies, and statistics. Data Engineering and Data Science are simply a form of applied statistics against specific data sets with the end goal being understanding. They can misrepresent the truth in the exact same way statistics at large can misrepresent the truth.

3.0.2 What responsibilities do analysts have when presenting results?

Analysts are responsible for understanding what their audience will take from the representation of the data they put forth because they are the only ones in the uniquely power position to understand the needs and biases of all the stakeholders at the table. These other stakeholders aren't stupid, but they aren't experts in data engineer and data science. As such, they don't know what they don't know. They will jump to conclusions based on sparse evidence. They will misinterpret data and think they understand. In short, they're human and it's part of the obligation of analysts to make sure that nobody is blind sided by the same issues that have plagued humanity for as long as we've had mathematical thought.

4 4. Analysis

4.0.1 Which variable appears most influential in explaining the outcome?

Sex, Island Name, and Mass were all very interesting in exploring the clustering I saw in the first mutli-plot graph against metrics. The collection of graphs displayed in “Categorical Comparison” section was the most useful single graph in that it displayed a lot at low resolution but facilitated a very intuitive few for the data in a thumbnail sized dashboard. This helped me understand what other values to investigate very quickly.

4.0.2 What insight would be most important to a decision maker?

Species distribution across islands was very interesting and would probably be used to inform any decision involving the cohort of penguins documented in this data set. The fact they all weren’t on all islands is very interesting. I also find the mass of the penguins interesting, especially when grouped by island, or by sex. What ended up delivering information the quickest was comparing by a pair of axes like mass and sex, or island and sex. Using scatter plots you understand the distribution easily and intuitively which was very useful for further investigation and would almost certainly be a useful phenomenon for a decision maker.

4.0.3 Which visualization best communicates your strongest finding and why?

The graphs in “Other Interesting Graphical Representations” section convey the species distribution across the island trivially. The graph in the “Distribution Chart” section clearly demonstrates the size distributions as well and how the species of penguins relate to each other by mass.

4.0.4 What limitations exist in your analysis?

I feel like the rigor of my numerival analysis was formulaic and poorly explained.

4.0.5 What additional data would improve your conclusions?

Sample size would help a lot at increasing confidence though this data was reasonable sized for a wildlife sampling. One thing that would help greatly would be the total population size of the various species such that we could better understand exactly how representational we could determine our mathematical analysis is. If it’s 1% of the population, we’d be able to trust it much more than if it was 0.01% of the population.